Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/21/2019
PI Name:	Blancaflor, Elison B. Ph.D.		
Project Title:	The Use of Microgravity Simulators for a Mechanistic Understanding of Cytoskeletal-Mediated Regulation of Root Growth		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	elison.b.blancaflor@nasa.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	321-501-5368
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	Utilization and Life Science Office		
PI Address 2:	Exploration and Research Technology Programs		
PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32955	**Congressional District:**	8
Comments:	NOTE: Dr. Blancaflor moved to NASA Kennedy Space Center in spring 2021; as a PI, his prior grants were for work at the Noble Foundation in Oklahoma.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	07/26/2018	**End Date:**	07/25/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	May 2019 report: Dr. Jing Wang -- Postdoctoral Fellow -- will be hired through this grant in July, 2019.		
COI Name (Institution):	Nakashima, Jin Ph.D. (Noble Research Institute)		
Grant/Contract No.:	80NSSC18K1462		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants will be a major component of advanced life support systems to enable NASA to realize its vision of long duration space exploration that will take humans to Mars and beyond. However, plants can only be effectively used for such systems if we have a deeper mechanistic understanding about how their development is regulated by the spaceflight environment. Recent "omics" studies have shown that plants reprogram their genetic circuitry when grown in the near weightless conditions of space. Genes involved in cell wall remodeling and oxidative stress are major gene categories that are differentially regulated by microgravity. Further, it has been shown that microgravity in space and simulated microgravity using clinostats or random positioning machines elicit roughly similar root skewing responses in the model plant Arabidopsis thaliana. Root skewing was more pronounced in mutants to a root-expressed vegetative actin isoform. Consistent with the enhanced root skewing of actin mutants in microgravity is the observation that other plant species such as maize (Zea mays L.) have stronger root directional growth responses on a clinostat when chemicals such as latrunculin B disrupt their actin cytoskeleton. Based on these observations, we will test the hypothesis that the actin cytoskeleton is a key player in dictating root growth directionality in microgravity and simulated microgravity by regulating the process of autotropic organ straightening. Moreover, we hypothesize that a set of cell wall cross-linking (i.e., class III peroxidases) and putative actin-regulatory proteins (i.e., HLB1 and MIN7/BEN1) mediate root hair elongation and directional root growth, respectively, under microgravity and simulated microgravity. Our hypotheses will be addressed through three specific aims including the analysis of cell wall composition/distribution in wild-type Arabidopsis roots, and detailed characterization of selected Arabidopsis cell wall and actin mutants subjected to microgravity simulation on a slow rotating clinostat and random position machine (aim 1 and 3). We will also use microgravity simulators to address the hypothesis that the plant hormone brassinolide regulates directional root growth in maize through its modulation of actin organization and dynamics (aim 2). Our research plan builds on successes of previous NASA-funded ground and spaceflight results, and outlines a comprehensive set of experiments including cell wall immunocytochemistry, computer-based root growth analyses, actin quantification, and genetics. The proposed work aligns with Plant Biology (PB) Element of the Space Biology Science Plan. Results obtained from plant experiments on microgravity simulators will address guiding questions linked to the decadal survey recommendations including the impact of gravity on plant growth, development and metabolism (PB-1), and molecular mechanisms underlying how plants sense and react to gravity (PB-5). We expect that the proposed research will provide us with basic knowledge that will guide the development of plant cultivars better adapted to the microgravity environment of spaceflight, an outcome that will prove beneficial for the design of plant habitats on future space colonies and for addressing great agricultural challenges on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal aims to gain a better mechanistic understanding of how plant roots function at the cellular and molecular levels. A primary focus of the proposed work is to ask how the microgravity environment modulates the directionality of root growth. Because roots are essential for the uptake of water and nutrients, knowledge gained from successful implementation of the proposed work will be crucial in guiding pre-breeding efforts aimed toward the development of crops with improved productivity through better and more efficient root systems. Thus, agriculture on Earth is one beneficial outcome of this project. Potential innovative technologies developed through this grant include lines of new model plants such as barrel medic (Medicago truncatula) and purple false brome (Brachypodium distachyon) expressing genetically-encoded sensors to the cytoskeleton and cytoplasmic calcium. We expect that these new plant lines will significantly advance basic plant biology research on Earth.
Task Progress:	Year one progress has come primarily from ground-based research conducted at our laboratory at the Noble Research Institute campus. As outlined in the proposal, this project has three specific aims, which are outlined below: Specific Aim 1. Immunocytochemistry of selected cell wall components in roots of Arabidopsis thaliana seedlings grown on a slowly-rotating clinostat and random positioning machine (RPM). Specific Aim 2. Characterize autonomic root straightening in maize roots in response to brassinolide (BL) - and latrunculin B (LatB) - induced actin disruption. Specific Aim 3. Evaluate root growth of selected Arabidopsis actin single and double mutants, and peroxidase mutants on microgravity simulators. Progress to date has been made mostly for aims 2 and 3. The experiments outlined in aim 2 have been completed. A peer-reviewed manuscript about the completed work in aim 2 will be submitted on July 31, 2019. Briefly, the experiments in aim 2 made use of a custom made 2-D clinostat to better understand how the actin cytoskeleton contributes to directional root growth. To address this question, we employed a combination of pharmacology and microscopic imaging of the filamentous-actin (F-actin) cytoskeleton in roots of maize (Zea mays) and Barrel Medic (Medicago truncatula). From previous NASA-funded research, we found that disruption of the F-actin cytoskeleton in roots using actin antagonists such as latrunculin B (LatB) caused roots to exhibit enhanced responses to gravity. Such effects were even more dramatic when plants were grown on a 2-D clinostat. Interestingly, the plant hormone brassinolide had roughly similar effects on root gravitropism as LatB, leading us to hypothesize that both compounds have similar effects on F-actin organization and dynamics. We found that brassinolide treatment modified both F-actin organization and global dynamics but such effects were not as dramatic as those observed for LatB-treated roots. High nanomolar (500 nM) to low micromolar concentrations (5 uM) brassinolide were needed to elicit quantifiable effects on F-actin organization in both chemically-fixed and living roots. For LatB, low nanomolar (25-100 nM) was sufficient to trigger both qualitative and quantitative effects on the root F-actin networks and root gravitropism. Taken together, our results from specific aim 2 indicate that brassinolide- and LatB-induced enhancement of root gravitropism can be explain by inhibited autotropic root straightening response, and that brassinolide affects this process by modifying F-actin organization and dynamics. We have also made substantial progress for the work outlined in aim 3. Most of our progress for aim 3 has been on the successful cloning of Arabidopsis thaliana root mutants identified via forward genetic screens. Mutants of particular interest are those that exhibited differential sensitivity to the actin-disrupting compound mentioned above (i.e., LatB). We are pleased to report that we have successfully cloned the disrupted genes in the hypersensitive to LatB 2 and 3 mutants (hlb2 and hlb3). Both are recessive mutants like our published hlb1 mutant from previous NASA-funded research. Like the HLB1 gene, HLB2 and HLB3 are proteins that link F-actin with the membrane system of plants. We plan to characterize root growth of hlb1, hlb2, and hlb3 mutants in more detail on the clinostat and random positioning

	machine (RPM) at the Kennedy Space Center (KSC)-Microgravity Simulation Support Laboratory (MSSL). Another exciting development regarding this aim is our successful cloning of the causal gene in one mutant that is resistant to LatB (i.e., LatB resistant – LBR1). Unlike the hlb mutants, Lbr1 is dominant and contains a mutation in the ACTIN7 gene. As implied from its name Lbr1 roots are able to grow despite the presence of high nanomolar concentrations of LatB. The ability of Lbr1 to grow on concentrations of LatB that severely inhibit wild type plants is consistent with observations that the actin cytoskeleton of Lbr1 remains intact even in the presence of LatB. We believe that the mutation in the Lbr1 mutant is affecting the ability of LatB to bind monomeric forms of actin, which subsequently makes LatB less toxic. Like the hlb mutants, we plan to evaluate root growth of Lbr1 on a 2-D clinostat and RPM. We hypothesize that like mutants to actin itself, some of these other mutants would exhibit enhanced or dampened root growth directional or skewing responses on microgravity simulators.
Bibliography Type:	Description: (Last Updated: 01/13/2026)
Abstracts for Journals and Proceedings	Kwon T, Sparks JA, Blancaflor EB. "The receptor-like kinase ERULUS regulates root hair tip growth through actin and calcium." Presented at the 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. Concurrent Technical Session: Plants 1 – Plants on Earth (I). Program and abstracts. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Nov-2018
Abstracts for Journals and Proceedings	Nakashima J, Pattathil S, Avci U, Sparks JA, Kwon T, Tang Y, Hahn MG, Blancaflor EB. "Changes in cell wall components of space-grown Arabidopsis roots uncovered by comparative glycomics." Presented at the 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. Concurrent Technical Session: Plants 2 – Plants Off Earth (I). Program and abstracts. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Nov-2018
Abstracts for Journals and Proceedings	Blancaflor EB. "ACAP-type ARF-GAPs as signaling hubs for maintenance of root hair polarity." Presented at the Plant Biology 2018, Montreal, Canada, July 14-18, 2018. Concurrent symposium on Cell Signaling. Abstracts. Plant Biology 2018, Montreal, Canada, July 14-18, 2018. , Jul-2018
Articles in Peer-reviewed Journals	Sun L, Ge Y, Sparks JA, Robinson ZT, Cheng X, Wen J, Blancaflor EB. "TDNAscan: A software to identify complete and truncated T-DNA insertions. " Front Genet. 2019 Jul 25;10:685. https://doi.org/10.3389/fgene.2019.00685 ; PubMed PMID: 31428129; PubMed Central PMCID: PMC6690219 , Jul-2019
Significant Media Coverage	Favre A. "Article on PI Blancaflor and his research, 'Space Exploration on Earth and How It Could Help Build Better Crops.' " Legacy: A Publication of The Noble Research Institute. 2018 Winter; p. 36-38. https://www.noble.org/news/publications/legacy/2018/winter/ , Dec-2018